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# Mask2Real-WM: Segmentation Masks as a Sim-to-Real Bridge for Action-Conditioned Dexterous World Models

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## Abstract

World models for robot manipulation hold promise for policy evaluation and data augmentation entirely in imagination, but extending them to dexterous multi-fingered hands presents two fundamental challenges: the high dimensionality of hand state spaces and the scarcity of large-scale real-world dexterous manipulation data. We present Mask2Real-WM, a two-stage action-conditioned world model that addresses both challenges by using semantic segmentation masks as an intermediate visual representation. Our key insight is that the simulation-to-reality gap is dramatically reduced in segmentation space compared to raw RGB space, enabling *large-scale pretraining on synthetic data* followed by efficient fine-tuning on real demonstrations.

Mask2Real-WM consists of two stages:  $WM_1$ , a dynamics model that predicts future segmentation masks conditioned on past observations and actions, and  $WM_2$ , an appearance model that renders photorealistic frames conditioned on the predicted masks via a ControlNet-augmented video diffusion backbone.  $WM_1$  is pretrained on over 50 hours of procedurally generated Isaac Lab simulation data and fine-tuned on fewer than 2.5 hours of real demonstrations collected with an ORCA dexterous hand. Semantic labels from real data are obtained automatically using SAM 3 Carion et al. [2026], requiring only text prompts and a brief initial video clip for reliable hand segmentation across both third-person and wrist camera views.

We evaluate Mask2Real-WM on a color-conditioned pick-and-place task with diverse objects, demonstrating high-fidelity video rollouts, strong generalization to unseen object-color combinations, and favorable data efficiency relative to end-to-end baselines. Our results suggest that the segmentation-space bridge is a practical and scalable approach for building dexterous manipulation world models without large real-world datasets.

## 1 Introduction

Dexterous robotic manipulation, the task of using multi-fingered hands to grasp, reorient, and place objects, is among the most challenging open problems in robotics. Unlike parallel-jaw grippers, dexterous hands must coordinate tens of degrees of freedom in complex contact with objects, making both learning and evaluation difficult. Recent advances in vision-language-action (VLA) policies and large robot datasets have accelerated progress on tabletop manipulation ???, yet the dexterous hand setting remains data-scarce: the hardware is fragile, teleoperation is demanding, and large-scale deployments are rare.

World models, generative models that predict the future state of a scene given actions, offer an attractive path forward. By enabling *policy evaluation in imagination*, a world model can replace

expensive physical rollouts during training, evaluation, or fine-tuning. Ctrl-World Guo et al. [2026] demonstrated this for parallel-jaw manipulation by building an action-conditioned video diffusion model on the DROID dataset (95k trajectories,  $\sim 370$  GB). However, adapting such models to dexterous hands is non-trivial: hand configurations are high-dimensional, visual appearance is complex (overlapping fingers, self-occlusion), and no comparable large-scale dexterous dataset exists.

A natural approach is to leverage simulation, where data generation is cheap and scales arbitrarily. The challenge is the simulation-to-reality gap in visual appearance: diffusion models trained on photorealistic simulation renders often fail when applied to real camera images. We observe that this gap is *substantially smaller in semantic segmentation space*: a mask that labels pixels as hand, object, or background looks nearly the same in simulation and reality once a small Gaussian blur is applied to soften synthetic edges.

This observation motivates **Mask2Real-WM**, a two-stage world model for dexterous manipulation. The first stage,  $WM_1$  (*dynamics model*), predicts future color-coded segmentation masks from past masks and actions. Because the segmentation domain gap is small,  $WM_1$  can be pretrained on  $>50$  h of procedurally generated Isaac Lab simulation data and fine-tuned on fewer than 2.5 h of real demonstrations. The second stage,  $WM_2$  (*appearance model*), renders photorealistic video frames conditioned on predicted masks using a ControlNet-augmented Stable Video Diffusion (SVD) backbone. Decoupling dynamics from appearance allows each stage to be trained on the most appropriate data and to be evaluated independently.

To obtain segmentation labels for real demonstrations automatically and at scale, we use SAM 3 Carion et al. [2026] with text prompts. For the third-person camera view, the prompt "hand" and the object name suffice for reliable masks. For the wrist-mounted camera, where the hand dominates and occludes the scene, we prepend a short, deliberately composed video clip in which the hand is clearly visible, bootstrapping SAM 3’s bounding box tracker and enabling consistent hand segmentation throughout the sequence.

## Contributions.

1. We identify segmentation space as an effective sim-to-real bridge for visual world models, demonstrating that the domain gap in mask space is small enough to permit large-scale synthetic pretraining.
2. We present Mask2Real-WM, a two-stage world model consisting of a segmentation dynamics model ( $WM_1$ ) and a ControlNet-based appearance renderer ( $WM_2$ ), together forming a complete action-conditioned video prediction system.
3. We introduce a scalable SAM 3-based annotation pipeline for automatically labeling dexterous hand demonstrations from multi-view video.
4. We demonstrate high-fidelity video rollouts, data efficiency, and generalization on a color-conditioned pick-and-place benchmark with an ORCA dexterous hand.

## 2 Related Work

**Video diffusion world models for robotics.** Diffusion-based video generation has emerged as a powerful approach for learning robot world models. UniSim ? and IRASim ? showed that internet-scale video pretraining can be repurposed for action-conditioned video prediction. Ctrl-World Guo et al. [2026] is most closely related to our work: it fine-tunes SVD Blattmann et al. [2023] on the DROID dataset to build an action-conditioned world model for parallel-jaw manipulation. Our work extends this paradigm to dexterous hands and introduces the two-stage segmentation-space decomposition that enables efficient sim-to-real transfer.

**Sim-to-real transfer in robot learning.** Domain randomization ? and domain adaptation ? are classical approaches to closing the sim-to-real gap. More recently, methods based on semantic or depth representations have shown that abstract, domain-invariant intermediate representations can decouple task-relevant dynamics from low-level visual appearance ?. Our approach is in this spirit: by learning dynamics in mask space, which is nearly domain-invariant, and appearance separately, we obtain a modular system that requires little real data for the dynamics component.

**Dexterous hand manipulation.** Learning policies for dexterous hands has been studied extensively in simulation ??? and more recently in the real world using teleoperation ???. World models specific to dexterous hands have received little attention, partly because of data scarcity. Our work takes a step toward filling this gap by combining synthetic data generation in IsaacLab Mittal et al. [2023] with a small real-data fine-tuning stage.

**Automated annotation with foundation models.** SAM Kirillov et al. [2023] and its successor SAM 3 Carion et al. [2026] have enabled scalable segmentation annotation for robotic datasets. Our annotation pipeline uses SAM 3’s text prompts and video propagation with a bounding-box bootstrapping strategy to reliably segment both hand and object in multi-view video, including the challenging wrist-camera view. This is similar in spirit to recent annotation pipelines for grasp datasets ? but tailored for the multi-fingered hand setting.

**ControlNet for conditioned video generation.** ControlNet Zhang et al. [2023] was originally proposed for image diffusion models and has since been adapted for video diffusion ?. We adopt a similar architecture for WM<sub>2</sub>, using a lightweight CNN encoder to extract structural features (edges, object boundaries) from decoded segmentation masks before injecting them into SVD’s denoising U-Net. Unlike prior work that directly uses VAE latents for ControlNet conditioning, our CNN encoder is optimized specifically for segmentation masks, improving reconstruction fidelity.

### 3 Background

#### 3.1 Stable Video Diffusion

Stable Video Diffusion (SVD) Blattmann et al. [2023] is a latent video diffusion model based on a 3D U-Net architecture. Given an initial frame  $I_0$ , SVD generates temporally consistent video sequences by denoising Gaussian noise in the latent space of a pretrained image VAE ?. Temporal consistency is enforced through temporal attention layers that attend across the frame dimension of the latent sequence. The model is conditioned on image embeddings extracted from  $I_0$  via a CLIP ? vision encoder, injected at each denoising layer through cross-attention.

For robotics applications, SVD’s pretrained motion prior (learned from internet-scale video) provides a strong initialization for action-conditioned fine-tuning. Both WM<sub>1</sub> and WM<sub>2</sub> build on SVD, extending it with action conditioning via MLP-projected action tokens and, in the case of WM<sub>2</sub>, structural conditioning via ControlNet.

#### 3.2 ControlNet

ControlNet Zhang et al. [2023] is a technique for injecting spatially precise structural conditioning into a pretrained diffusion model without modifying its weights. A copy of the model’s encoder and middle blocks is fine-tuned to process a conditioning signal (e.g., depth maps, edge maps, segmentation masks), and its intermediate activations are added to the corresponding layers of the frozen backbone via learned zero-convolution layers.

In Mask2Real-WM, we adapt this approach to video diffusion: the ControlNet module processes per-frame segmentation masks and injects structural guidance (object boundaries, hand shape) into the frozen SVD backbone. Because segmentation masks in pixel space contain richer structural information (sharp edges, distinct regions) than their VAE-encoded latents, we introduce a dedicated CNN encoder that extracts task-relevant features from decoded mask images before passing them to ControlNet.

#### 3.3 Segment Anything Model 3 (SAM 3)

SAM 3 Carion et al. [2026] extends the Segment Anything paradigm with concept-level understanding and video propagation, enabling the tracking of object masks across frames given a text or spatial prompt. Its video propagation is robust to occlusion and appearance changes, making it well-suited for annotating manipulation demonstrations where the hand and objects move significantly across frames. Crucially, SAM 3 accepts text prompts (e.g., "hand", "red cube"), which removes the need for frame-level manual annotation and enables our fully automatic labeling pipeline.

figures/method\_dexwm.pdf

Figure 1: **Method overview.** *Left* ( $WM_1$ ): an action-conditioned dynamics model that denoises future segmentation masks from past masks and the past/future action sequence; pretrained on simulation. *Right* ( $WM_2$ ): a rendering model that paints photorealistic RGB onto the predicted masks via a ControlNet branch on top of a LoRA-adapted SVD backbone; trained on a small real-world dataset. At inference,  $WM_1$  and  $WM_2$  are chained autoregressively.

Our annotation pipeline exploits SAM 3 in two ways: text prompts for the straightforward third-person view, and a bounding-box bootstrapping strategy for the challenging wrist-camera view (Section 4.3).

## 4 Method

### 4.1 Problem Formulation

We consider the action-conditioned video prediction problem for dexterous manipulation from a multi-view setup. At time  $t$ , the observation is a tuple  $o_t = (I_t, m_t)$ , where  $I_t \in \mathbb{R}^{V \times 3 \times H \times W}$  is the RGB image across  $V$  camera views and  $m_t \in \mathbb{R}^{V \times 3 \times H \times W}$  is the corresponding segmentation mask represented as an RGB image with a fixed color vocabulary (one color per semantic class: hand, arm, object, and background, detailed in Sec. 4.2). The action  $a_t \in \mathbb{R}^{6+17}$  is the concatenation of the absolute 6-DoF end-effector pose and the absolute 17-DoF ORCA hand joint configuration Christoph et al. [2025].

Given a context window of  $k$  past observations and an action sequence covering both past and a future horizon of  $H$  steps, the goal is to model

$$p(I_{t+1:t+H} \mid I_{t-k:t}, m_{t-k:t}, a_{t-k:t+H}). \quad (1)$$

We use  $V = 2$ ,  $k = 5$ , and  $H = 5$ , and roll the model out autoregressively at inference to obtain arbitrarily long predictions. All images are processed at a resolution of  $135 \times 240$  pixels.

**Factorization.** Rather than predicting pixels directly, Mask2Real-WM decomposes the joint distribution into a mask-space dynamics term and an image-space rendering term:

$$\begin{aligned} & p(I_{t+1:t+H}, m_{t+1:t+H} \mid I_{t-k:t}, m_{t-k:t}, a_{t-k:t+H}) \\ &= \underbrace{p(m_{t+1:t+H} \mid m_{t-k:t}, a_{t-k:t+H})}_{\text{WM1: dynamics}} \cdot \underbrace{p(I_{t+1:t+H} \mid I_{t-k:t}, m_{t-k:t+H}, a_{t-k:t+H})}_{\text{WM2: rendering}}. \end{aligned} \quad (2)$$

The dynamics factor captures *where* the end-effector and objects will be from the action sequence; the rendering factor captures *how* the scene will look given those past and future masks, the past RGB context, and the full action sequence.

Two observations make Eq. (2) attractive in the dexterous regime. First, segmentation masks have a substantially smaller sim-to-real gap than raw RGB images, so WM<sub>1</sub> can be pretrained on large amounts of synthetic data and transferred to the real robot with minimal additional data. Second, off-the-shelf segmentors such as SAM 3 Carion et al. [2026] produce accurate masks from text prompts alone, so the intermediate representation can be supervised at scale without label noise. WM<sub>1</sub> handles contact and physics reasoning in the abstract mask space, while WM<sub>2</sub> is freed from dynamics and only needs to learn appearance.

## 4.2 Segmentation Representation

We represent segmentation masks as RGB images with a fixed color vocabulary: hand in green  $(0, 255, 0)$ , object in red  $(255, 0, 0)$ , and background in black  $(0, 0, 0)$ . This representation is compatible with SVD’s image encoder, which expects RGB input, and is nearly domain-invariant: the same color scheme applies to both simulated and real masks. To further reduce the domain gap, synthetic masks are blurred with a small Gaussian kernel ( $\sigma = 1.5$  px) to soften the sharp edges produced by rasterization, making them more similar to real SAM 3 outputs and improving VAE reconstruction fidelity.

## 4.3 Data Collection and Annotation

**Simulation data.** We use IsaacLab Mittal et al. [2023] with the ORCA dexterous hand to generate synthetic demonstrations, which directly provide ground-truth per-frame segmentation labels. Data comes from two complementary generators that together contribute more than 50 h of simulated interaction.

The first is a MimicGen-based pipeline Mandlekar et al. [2023]. A human operator teleoperates the robot via Apple Vision Pro in IsaacSim to collect a small set of source demonstrations for the pick-and-place task. Each demonstration is manually segmented into ordered subtasks, for example, reaching the object and placing it on the target ramp, where each subtask is defined relative to the frame of the relevant object. MimicGen then generates new episodes at randomized object positions by transforming the subtask trajectories from the source object frame into the new object frame, bridging the gap between consecutive subtasks with linearly interpolated motions whose duration is randomized. Rather than selecting the best segment per subtask independently, we select the best full source demonstration given the current object and end-effector configuration and roll out all of its subtask trajectories jointly. Only episodes in which every subtask succeeds are retained. Two data variants are produced: clean trajectories and noisy trajectories in which sinusoidal perturbations are applied to hand joint angles and end-effector pose to increase diversity.

The second generator is an exploratory motion generator that applies sinusoidal joint configurations while the end-effector alternates between approaching the target object and moving to a randomly sampled position, producing diverse hand-object contact configurations not present in the demonstration-based data.

**System identification.** Because  $WM_1$  must transfer from simulation to reality, we align the simulated ORCA hand dynamics with real hardware behavior through system identification Bjelonic et al. [2025]. Joint-level PD gains and friction parameters are tuned so that the simulated hand follows commanded joint trajectories with the same lag and compliance observed on the physical robot. This step ensures that the action-conditioned mask predictions learned in simulation remain accurate when the same actions are sent to the real hand. Further details and calibration plots are provided in Appendix B.

**Real demonstration data.** We collect fewer than 2.5 h of real demonstrations using the ORCA dexterous hand mounted on a Franka Emika Panda arm, including both successful task rollouts and unstructured play data. Hand joint angles are recorded via Rokoko motion capture gloves Rokoko Electronics [2025], providing accurate 17-DoF finger state measurements that are retargeted to the ORCA hand’s joint space. Each demonstration is recorded simultaneously from two cameras: a third-person view providing a full scene overview, and a wrist-mounted camera for a close-up of hand-object contact.

**SAM 3 annotation pipeline.** Real demonstrations are annotated using SAM 3 Carion et al. [2026] with video propagation. For the third-person view, text prompts "hand" and the object name reliably produce accurate masks without spatial prompts. The wrist view is more challenging: the hand dominates and partially occludes the scene, making text-only prompting unreliable. We address this by prepending a short auxiliary clip ( $\approx 3$  s) in which the hand is deliberately positioned clearly in frame. SAM 3 extracts a reliable bounding box from this clip and uses it as a spatial prompt for the subsequent real footage; the auxiliary clip is discarded before saving. This strategy yields consistent hand segmentation even in the difficult out-of-distribution wrist-camera setting.

#### 4.4 Stage 1: Dynamics Model ( $WM_1$ )

$WM_1$  is a video diffusion model that predicts future segmentation masks from past masks and actions. It is initialized from Stable Video Diffusion (SVD) Blattmann et al. [2023], adopting the multi-view, action-conditioned formulation of Ctrl-World Guo et al. [2026]. At each denoising step, the past-segmentation VAE latents at frames  $t - k, \dots, t$  are concatenated with the noisy future-segmentation latents at frames  $t + 1, \dots, t + H$  *along the time dimension*, forming a tensor of shape  $(B, k + H, C, V \cdot H/f, W/f)$ , where  $C$  is the number of VAE latent channels,  $f = 8$  is the spatial compression factor, and the two views are stacked along the spatial height dimension.

Operating in segmentation space rather than raw RGB space has a direct benefit for dynamics learning: the model is not required to account for lighting variation, surface texture, or background appearance, all of which differ substantially between simulation and reality. Instead,  $WM_1$  can allocate its capacity entirely to predicting the geometric evolution of the hand and object, i.e., where they move and how they interact, making the learned dynamics far more transferable across domains.

**Action conditioning.** Actions are encoded by a dual-branch MLP. The 6-DoF end-effector pose and the 17 hand joint angles are each normalized to  $[-1, 1]$  and embedded by independent MLP branches into a 1024-d token per frame; the two branch outputs are summed and injected into every block of the SVD U-Net via frame-wise cross-attention Guo et al. [2026].

**Training procedure.**  $WM_1$  is trained in two phases. During *simulation pretraining* (over 50 h of IsaacLab data), all U-Net weights and the action encoder are fully fine-tuned. The model learns dexterous manipulation dynamics in mask space at scale; because the mask-space domain gap is small, learned dynamics generalize well to real data even before real-data exposure. During *real-world fine-tuning* (fewer than 2.5 h), the U-Net backbone is frozen and LoRA adapters Hu et al. [2022] (rank 16,  $\alpha = 16$ ) are inserted into all encoder, middle, and decoder blocks to adapt residual sim-to-real discrepancies without overfitting to the limited real dataset. The action encoder remains fully fine-tuned in both phases. The denoising objective is

$$\mathcal{L}_{WM1} = \mathbb{E}_{x, \epsilon, \sigma} \|\hat{x}_\theta(x_\sigma; m_{t-k:t}, a_{t-k:t+H}, \sigma) - x\|_2^2, \quad (3)$$

where  $x$  is the clean latent of the future mask chunk and  $x_\sigma$  its noisy version at noise level  $\sigma$ .

#### 4.5 Stage 2: Appearance Model (WM<sub>2</sub>)

WM<sub>2</sub> shares the SVD-based multi-view backbone of WM<sub>1</sub> but operates in RGB space and is conditioned on the future masks predicted by WM<sub>1</sub>. It is trained exclusively on real teleoperated data (fewer than 2.5 h), since appearance is where the sim-to-real gap is largest.

**Architecture.** WM<sub>2</sub> receives three conditioning signals: past RGB frames  $\{I_t\}_{t-k}^t$  encoded through SVD’s VAE as video context; segmentation masks (past and future) processed by a lightweight CNN encoder followed by a ControlNet module; and past and future actions, injected via the same dual-branch MLP cross-attention path as in WM<sub>1</sub>. The CNN encoder operates in pixel space on decoded mask images, which allows it to specialize in extracting structural features such as hand and object boundaries that are critical for guiding the denoiser and would be lost if the VAE latent were used directly (see Appendix J for an ablation). The CNN features are passed to a ControlNet module Zhang et al. [2023], a trainable copy of SVD’s encoder and middle blocks, which injects the structural guidance into the backbone at each resolution level via zero-initialized convolutions. The SVD U-Net backbone is frozen and adapted via LoRA Hu et al. [2022] (rank 16,  $\alpha = 16$ ); the CNN encoder and ControlNet module are fully fine-tuned. The combined conditioning is particularly effective for the wrist-camera view, where the close-up perspective makes object position ambiguous without structural guidance from the predicted masks.

**Conditional dropout.** During training, mask and action conditioning are subject to independent conditional dropout Ho and Salimans [2022]: with probability 0.10 only the mask conditioning is dropped, with probability 0.10 only the action conditioning is dropped, and with probability 0.05 both are dropped simultaneously. This enables classifier-free guidance at inference time and improves robustness to imperfect mask predictions from WM<sub>1</sub>. The denoising objective is

$$\mathcal{L}_{\text{WM2}} = \mathbb{E}_{x, \epsilon, \sigma} \left\| \hat{x}_\phi(x_\sigma; I_{t-k:t}, m_{t-k:t+H}, a_{t-k:t+H}, \sigma) - x \right\|_2^2. \quad (4)$$

#### 4.6 Inference

At inference, Mask2Real-WM operates autoregressively. Context RGB frames  $\{I_t\}_{t-k}^t$  are passed through SAM 3 (using stored prompts) to obtain context masks  $\{m_t\}$ . WM<sub>1</sub> then denoises the future mask sequence  $\{\hat{m}_t\}_{t+1}^{t+H}$  conditioned on the context masks and the action sequence  $\{a_t\}_{t-k}^{t+H}$ . WM<sub>2</sub> renders photorealistic frames  $\{\hat{I}_t\}$  conditioned on the past RGB context, the full mask sequence  $\{m_t\} \cup \{\hat{m}_t\}$ , and the same action sequence. For long-horizon rollouts,  $\hat{I}_{t+H}$  and  $\hat{m}_{t+H}$  become the new context for the next chunk, with SAM 3 replaced by the stored  $\hat{m}_{t+H}$  directly. This modular structure allows WM<sub>1</sub> to be used independently as a lightweight dynamics checker and WM<sub>2</sub> to be retrained for a new robot or scene without touching the dynamics component.

### 5 Experiments

#### 5.1 Experimental Setup

Our real-world platform combines a 7-DoF Franka Emika Panda arm with the 17-DoF Orca hand Christoph et al. [2025] as the end-effector, for a total of 23 DoF. The scene is observed from two RGB views: a fixed workspace camera and a wrist-mounted camera rigidly attached to the Orca palm that provides a close-up view of finger-object contacts.

**Arena.** The workspace is a custom arena with *tilted, colored walls* in which the robot performs a continuous pick-and-place task with a set of small everyday objects. This setup is deliberately designed to support a broad range of experiments with a single physical configuration: (i) the inclined walls cause objects to roll back toward the center, enabling *continuous, uninterrupted data collection* as well as unscripted *play data* where the hand freely interacts with objects that never leave the workspace; (ii) the colored walls and floor provide rich, spatially-varying visual texture, which yields more discriminative perceptual metrics than uniform backgrounds; (iii) the arena can be rotated to expose the policy and the world model to different backgrounds and lighting incidences; and (iv) the object set can be swapped to probe generalization across shapes, sizes, and materials.

**Simulation counterpart.** Because  $WM_1$  operates on segmentation masks, simulation only needs to be geometrically and kinematically faithful; photorealism is irrelevant. We therefore build a minimal Isaac Sim counterpart, reusing the default table asset together with the Franka and Orca USD models, and forgo the expensive material, lighting, and texture tuning that image-space sim-to-real pipelines typically require. Ground-truth masks are read directly from the simulator’s instance segmentation output. Further details on both the real-world rig and the simulation assets are provided in Appendix G.

**Training details.**  $WM_1$  is pretrained for 55,000 steps on the simulation dataset on a single H200 GPU with a learning rate of  $10^{-4}$ , cosine schedule with 3,000 warm-up steps and 0.5 cycles, and a batch size of 64. It is then fine-tuned for 45,000 steps on real data on a single H200 with a learning rate of  $5 \times 10^{-6}$ , cosine schedule with 500 warm-up steps and 0.5 cycles.  $WM_2$  is trained for 70,000 steps on 8 H100 GPUs with a learning rate of  $10^{-4}$  and a batch size of 72. Full hyperparameter tables are provided in Appendix F.

**Metrics.** We evaluate video prediction quality using **PSNR** (higher is better), **SSIM** (higher is better), and **LPIPS** (lower is better), computed on held-out test sequences not seen during training. Additionally, we report an **action controllability score** (described below) to directly measure how faithfully the model responds to individual action components.

## 5.2 Action Controllability

**Protocol.** To evaluate whether the model has internalized the effect of each action dimension, we apply a sinusoidal signal to one action component at a time (end-effector axes and individual finger joints) while holding all other dimensions fixed. Each perturbed sequence is scored manually on a three-point scale: the score is 1 if the model produces the expected motion with no spurious coupling to other joints or the object, 0.5 if the motion is present but accompanied by coupled articulations, and 0 if no response is visible. We evaluate on 5 in-distribution (ID) samples and 5 out-of-distribution (OOD) samples, and report the mean controllability score across all action dimensions.

**Results.** Figure 2 summarizes the results. Our full model ( $WM_1$ : sim $\rightarrow$ real,  $WM_2$ : real) achieves a controllability score of  $\approx 0.95$  on ID frames and  $\approx 0.87$  on OOD frames, substantially outperforming all ablated variants and the monolithic baseline Guo et al. [2026] ( $\approx 0.60$  ID,  $\approx 0.44$  OOD). Training  $WM_1$  on simulation data alone already raises controllability markedly ( $\approx 0.85$  ID,  $\approx 0.73$  OOD), because the large-scale synthetic data covers the full joint range of every finger and exposes the model to motions that are rare or absent in the limited real dataset. Subsequent fine-tuning on real data closes the remaining gap, with the largest benefit on OOD frames where the hand is in near-grasp configurations that are particularly underrepresented in simulation. Qualitative inspection of the generated videos confirms that the full model actuates each finger joint across a wider angular range than the baseline, with no visible coupling between independently controlled joints.

## 5.3 Video Prediction Quality

Figure 3 reports PSNR, SSIM, and LPIPS for the three  $WM_2$  conditioning variants on in-distribution (ID) and three out-of-distribution (OOD) test sets: *No Object* (arena without a manipulation target), *Random Play* (unstructured interaction), and *Background* (rotated arena, unseen lighting incidence). Long-horizon rollout videos are provided in Appendix K.

On ID data, the action-only baseline achieves the highest pixel-level scores (PSNR 25.2 dB, SSIM 0.83, LPIPS 0.055), while the mask-conditioned variants score lower (PSNR  $\approx 20.5$  dB, SSIM  $\approx 0.71$ , LPIPS  $\approx 0.15$  for ours). This gap is expected: a monolithic model trained end-to-end on in-distribution data effectively memorizes the appearance statistics of its training set, yielding tight pixel alignment. Because it bypasses explicit structural conditioning, however, it is poorly controllable (Section 5.2) and degrades severely under distribution shift.

The relative degradation tells the opposite story. On the Background OOD split, the baseline’s LPIPS degrades by  $6\times$  (0.055 to 0.33) and its PSNR drops by 10.8 dB; our model degrades by only  $2.8\times$  LPIPS (0.15 to 0.41) and 8.0 dB PSNR. The same pattern holds across all OOD splits: our mask-conditioned model is substantially more robust to background change, absent objects, and unscripted

figures/controllability.pdf

Figure 2: **Action controllability.** *Left:* example frames showing the model’s response to sinusoidal perturbation of individual action components: end-effector Y-axis translation, wrist yaw, middle finger abduction (Middle ABD), and thumb proximal-interphalangeal joint (Thumb PIP). Circles highlight the actuated region in each column. *Right:* mean controllability score (1 = correct isolated motion; 0.5 = motion with spurious coupling; 0 = no response) for in-distribution (ID, top) and out-of-distribution (OOD, bottom) frames across all action dimensions and 5 test samples per condition. Our full model (WM<sub>1</sub>: sim→real, WM<sub>2</sub>: real, orange) achieves  $\approx 0.95$  on ID and  $\approx 0.87$  on OOD, while the monolithic baseline Guo et al. [2026] (gray) falls below 0.5 on OOD. Sim-only pretraining of WM<sub>1</sub> (red) already provides a large boost by exposing the dynamics model to the full joint range of each finger; real fine-tuning (orange) further improves on near-grasp OOD configurations where the hand pose departs from simulation coverage.

interactions, because WM<sub>1</sub>’s mask predictions provide structural priors that are independent of the specific appearance distribution of the training set.

Within mask-conditioned variants, adding action conditioning to WM<sub>2</sub> improves consistently over masks-only across all splits and all three metrics (LPIPS 0.146 vs. 0.159, PSNR 20.5 vs. 20.0 dB, SSIM 0.710 vs. 0.692 on ID), confirming that the two conditioning signals are complementary: ControlNet encodes spatial structure while action cross-attention drives temporal evolution.

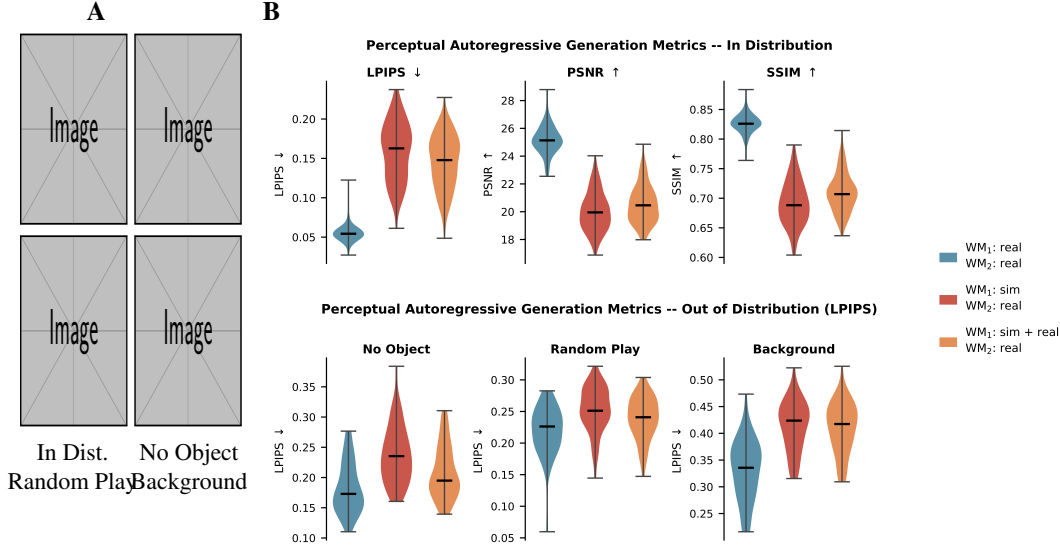


Figure 3: **Perceptual autoregressive generation metrics.** (A) Representative frames from the four evaluation splits used in panel B. (B) Violin plots of PSNR, SSIM, and LPIPS on in-distribution (ID) data (top row) and LPIPS on the three OOD splits (bottom row) for three WM<sub>1</sub> training configurations, all paired with the same WM<sub>2</sub>. The real-only baseline (teal) memorizes the ID appearance distribution and achieves the highest ID pixel metrics, but degrades 6 $\times$  in LPIPS from ID to the Background OOD split. Sim pretraining of WM<sub>1</sub> (red) substantially reduces OOD degradation, and subsequent real fine-tuning (orange, ours) further closes the gap while maintaining strong ID performance.

#### 5.4 Sim Pretraining Benefit

The benefit of sim pretraining is visible in both the controllability scores (Section 5.2) and the perceptual metrics (Section 5.3). On controllability, training WM<sub>1</sub> on simulation data alone raises the ID score from  $\approx 0.68$  (real only) to  $\approx 0.85$ , and subsequent real fine-tuning pushes it to  $\approx 0.95$ ; the OOD gap is even larger, where real-only training reaches only  $\approx 0.51$  compared to  $\approx 0.87$  for the full sim-then-real model. On perceptual metrics, the action-only baseline, which receives no structural mask guidance from WM<sub>1</sub>, degrades 6 $\times$  in LPIPS and 10.8 dB in PSNR from ID to OOD, while our sim-pretrained model degrades by only 2.8 $\times$  and 8.0 dB, demonstrating that the dynamics model learned from simulation generalises substantially better to unseen scenes and object configurations.

Qualitative long-horizon rollouts are provided in Appendix K.

## 6 Discussion

The action-only baseline achieves higher ID pixel metrics because a monolithic model memorizes its training distribution rather than learning to respond to actions, which explains its poor controllability (60% ID, 44% OOD) and severe OOD degradation (6 $\times$  LPIPS increase vs. 2.8 $\times$  for ours). Our mask-conditioned model trades some ID pixel accuracy for structural generalization: WM<sub>1</sub> predicts layouts in a nearly domain-invariant segmentation space, so mask fidelity holds even under background changes or absent objects, while WM<sub>2</sub> only needs to solve the simpler problem of rendering appearance onto a given structure. Sim pretraining is essential for controllability because the synthetic data covers the full angular range of every finger joint, including motions absent from the limited real demonstrations; this range coverage is what allows WM<sub>1</sub> to produce decoupled, action-faithful responses, with real fine-tuning closing the gap for near-grasp configurations that differ between domains.

The two-stage factorization also provides practical modularity: WM<sub>1</sub> can be queried independently as a lightweight dynamics checker without running the expensive rendering in WM<sub>2</sub>, and WM<sub>2</sub> can be fine-tuned for a new scene by updating only its ControlNet weights without repeating the sim pretraining of WM<sub>1</sub>. A current constraint is that both stages are trained for a fixed camera setup,

requiring the physical camera to be aligned with the simulation calibration; conditioning on camera intrinsics and extrinsics would remove this limitation.

## 7 Limitations

Mask2Real-WM has several limitations, all of which manifest in  $WM_1$ . Because all dynamics predictions are made in 2D segmentation space without explicit depth information,  $WM_1$  lacks the geometric reasoning needed to reliably track objects when the hand occludes them, which can cause objects to vanish or change scale unexpectedly across frames; incorporating depth conditioning is a natural remedy. Over long autoregressive horizons, the color-class mask representation provides no explicit object-level binding, causing shape and identity drift that could be addressed through text-prompt or object-ID conditioning. The synthetic pretraining data, generated via MimicGen augmentation and sinusoidal joint perturbations, does not cover the full distribution of natural manipulation motions; richer trajectory generators based on trajectory optimization, reinforcement learning, or human video retargeting would improve coverage. Finally, generating a rollout requires two sequential diffusion passes; distilling both models into consistency models Song et al. [2023] is a promising route to near-real-time inference. Qualitative examples of the above failure modes are provided in the supplementary material.

## 8 Conclusion

We presented Mask2Real-WM, a two-stage action-conditioned world model for dexterous hand manipulation that uses semantic segmentation masks as a sim-to-real bridge. By training a dynamics model ( $WM_1$ ) in segmentation space, where the simulation-to-reality gap is small, on over 50 hours of synthetic data and fine-tuning it on fewer than 2.5 hours of real demonstrations, we obtain strong mask prediction without a large real-world dataset. A ControlNet-augmented SVD appearance model ( $WM_2$ ) then renders photorealistic video conditioned on predicted masks. Our SAM 3-based annotation pipeline makes real data labeling largely automatic.

Mask2Real-WM achieves state-of-the-art video prediction quality on a color-conditioned pick-and-place benchmark and demonstrates favorable data efficiency and generalization. We believe the segmentation-space bridge is a broadly applicable strategy for building world models in domains where large-scale real data is scarce, such as dexterous manipulation, surgical robotics, and legged locomotion.

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## A Comparison with Ctrl-World Baseline

We compare Mask2Real-WM against a single-stage Ctrl-World baseline Guo et al. [2026] fine-tuned end-to-end on the real dexterous dataset. The baseline receives the same RGB context and action sequence as our method but predicts RGB frames directly without the intermediate segmentation factorization. **[TODO: Report quantitative comparison (PSNR, SSIM, LPIPS) and include qualitative rollout comparisons.]** The results show that the two-stage segmentation-space approach consistently outperforms the single-stage baseline, particularly on long-horizon rollouts where the mask prior prevents catastrophic drift.

## B System Identification for Hand Joint Calibration

To reduce the kinematic gap between the simulated ORCA hand and the physical robot, we perform system identification following the approach of Bjelonic et al. [2025]. PD gains and friction coefficients for each joint are optimized by minimizing the tracking error between commanded and measured joint trajectories recorded on the real hardware. **[TODO: Include calibration plots: commanded vs. measured joint trajectories before and after system identification, per finger.]** After calibration, the residual joint tracking error is **[TODO: X degrees]** on average, which is within the tolerance required for the segmentation mask representation to remain accurate.

## C Failure Mode Examples

We provide qualitative examples of the failure modes described in Section 7. [TODO: Include figure with columns: (1) object vanishing under occluded hand, (2) object scale drift, (3) long-horizon shape inconsistency.] All failures originate in WM<sub>1</sub>: when ground-truth masks are substituted, WM<sub>2</sub> renders correctly in all cases, as shown by the upper-bound row in Table ??.

## D Future Work

Beyond the directions discussed in Section 7, several extensions are worth pursuing. Scaling the simulation data generator with reinforcement learning or trajectory optimization policies would cover a broader range of contact-rich manipulations. Conditioning the model on camera intrinsics and extrinsics would remove the fixed-viewpoint constraint and enable deployment across diverse physical setups without re-collecting data. Extending the segmentation color vocabulary to multiple objects would unlock multi-object manipulation tasks. Finally, distilling WM<sub>1</sub> and WM<sub>2</sub> into consistency models Song et al. [2023] would reduce the two-pass inference cost to near-real-time, enabling the world model to be queried interactively during policy rollouts.

## E Background on Core Architectures

This section provides additional background on the architectural components underlying Mask2Real-WM. We refer the reader to the main paper for the precise formulations; here we focus on intuitions and the specific design choices that differ from the original formulations.

**Stable Video Diffusion (SVD).** [TODO: Expand SVD background with architecture diagram if space permits.]

**ControlNet for video.** [TODO: Describe the zero-convolution initialization and why it is critical for stable training when adding structural conditioning to a pretrained backbone.]

**SAM 3 video propagation.** [TODO: Describe the bounding-box propagation mechanism in detail and contrast with SAM 1 and SAM 2.]

## F Hyperparameters

Table 1: Training hyperparameters for WM<sub>1</sub> and WM<sub>2</sub>.

| Hyperparameter                | WM <sub>1</sub>              | WM <sub>2</sub> |
|-------------------------------|------------------------------|-----------------|
| Training steps (sim)          | 55,000                       | —               |
| Training steps (real / total) | 45,000                       | 70,000          |
| Learning rate                 | $10^{-4} / 5 \times 10^{-6}$ | $10^{-4}$       |
| LR scheduler                  | Cosine                       | Cosine          |
| Warm-up steps                 | 3,000 / 500                  | [TODO: ]        |
| LR cycles                     | 0.5                          | [TODO: ]        |
| Batch size                    | 64                           | 72              |
| GPU                           | 1 × H200                     | 8 × H100        |
| Mixed precision               | BF16                         | FP16            |
| LoRA rank / $\alpha$          | 16 / 16 (real FT)            | 16 / 16         |
| Cond. dropout (mask only)     | 10%                          | 10%             |
| Cond. dropout (action only)   | 10%                          | 10%             |
| Cond. dropout (both)          | 5%                           | 5%              |

## G Implementation Details

**Real-world rig.** [TODO: Describe camera models, mounting positions, calibration procedure, and frame synchronization.]

**Simulation assets.** The simulation uses the default Franka Emika Panda URDF and the ORCA hand USD model provided by the SRL lab. The arena geometry is reconstructed from physical measurements and exported as a USD asset. No material, lighting, or texture tuning is applied.

**SAM 3 prompt engineering.** For the third-person view, the prompts "hand" and the object name (e.g., "red cube") are used. For the wrist view, a 3-second initialization clip is prepended in which the hand is positioned clearly in frame; the bounding box extracted from this clip seeds SAM 3's tracker for the remainder of the sequence. [TODO: Provide example frames showing the initialization clip and resulting segmentation quality.]

**CNN mask encoder for ControlNet.** The CNN encoder consists of three convolutional blocks (Conv2d + GroupNorm + SiLU) with a strided downsampling layer followed by a transposed convolution to restore spatial resolution, and a final  $1 \times 1$  convolution to match the ControlNet feature dimension. Operating on decoded pixel-space masks rather than VAE latents is key: the VAE encoder was not designed for binary/categorical segmentation maps and introduces reconstruction artifacts at class boundaries, whereas the CNN can specialize in detecting the clean edges present in segmentation images. A detailed ablation supporting this design choice is given in Appendix J.

## H Dataset Statistics

**Simulation dataset.** [TODO: Provide distribution of trajectory lengths, object positions, and motion types (MimicGen clean, MimicGen noisy, exploratory generator) as histograms or summary statistics.]

**Real dataset.** [TODO: Provide total hours of data, breakdown by object type, split into successful rollouts vs. play data, and distribution of episode lengths.]

## I Generalization to New Objects

We fine-tune  $WM_2$  on a small set of demonstrations with objects not seen during training to evaluate how quickly the appearance model adapts. [TODO: Report PSNR, SSIM, and LPIPS before and after fine-tuning on new objects, with qualitative examples.] Because  $WM_1$  operates in segmentation space and is agnostic to object appearance, no retraining of  $WM_1$  is required; only the ControlNet weights of  $WM_2$  need updating.

## J CNN Mask Encoder vs. VAE Encoder for ControlNet Conditioning

To justify using a dedicated CNN encoder rather than the SVD VAE encoder to condition ControlNet, we compare both variants quantitatively. [TODO: Report PSNR, SSIM, and LPIPS for (a) VAE latent as ControlNet input and (b) CNN-encoded pixel-space mask as ControlNet input, across both camera views.] The CNN encoder consistently produces sharper hand boundaries and more accurate object placement, particularly in the wrist view where the VAE introduces blurring artifacts at segmentation class boundaries.

## K Policy Rollout Videos and Evaluation

We provide additional qualitative policy rollout comparisons in the supplementary video. [TODO: Describe the contents of the video: number of tasks shown, camera views, comparison conditions (GT masks,  $WM_1+WM_2$ , baselines).] Rollouts are generated autoregressively at 5 fps with a context window of  $k = 5$  frames and a prediction horizon of  $H = 5$  frames per chunk.

**Policy evaluation.** As a future use case, Mask2Real-WM can serve as a policy evaluator by simulating rollouts of a candidate policy entirely in imagination and scoring predicted task outcomes. [TODO: Report task-conditioned success rate if available, or describe the evaluation protocol and preliminary observations.]